REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Describing Rotational Motion Practice Questions

Grade 11 Advanced | Module 8 Lesson 1 | Describing Rotational Motion

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: $\theta = s/r$; $\omega_{avg} = \Delta\theta/\Delta t$; $\alpha_{avg} = \Delta\omega/\Delta t$; $v = r\omega$; $a_t = r\alpha$; $\omega = 2\pi f$; 1 revolution = 2π rad.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 11 book, printed pages 3–6.

P01. One complete revolution equals:

A. π radians B. 2π radians
C. 180 radians D. 1 radian

P02. Points on the same rigid wheel share the same:

A. Tangential speed at every radius B. Distance travelled in every rotation
C. Angular velocity D. Tangential displacement magnitude

P03. For fixed angular velocity, tangential speed is:

A. Inversely proportional to radius B. Independent of radius always
C. Equal to radius without a time factor D. Proportional to radius

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P04. Angular acceleration is the rate of change of:

A. Angular velocity B. Radius
C. Mass D. Moment of inertia only

P05. A radian is related to arc length s and radius r by:

A. $\theta = sr$ B. $\theta = s/r$
C. $\theta = r/s$ for all angles D. $\theta = s + r$

P06. Convert 90° to radians.

A. 3.142 rad B. 0.7854 rad
C. 1.571 rad D. -1.571 rad

P07. A wheel turns through 8 rad in 4 s. Calculate its average angular velocity.

A. 4 rad/s B. 1 rad/s
C. -2 rad/s D. 2 rad/s

P08. A point is 0.2 m from a rotating axle with angular velocity 5.0 rad/s. Calculate its tangential speed.

A. 1 m/s B. 2 m/s
C. 0.5 m/s D. -1 m/s

P09. A wheel's angular velocity changes from 2 to 14 rad/s in 6 s. Find its average angular acceleration.

A. 4 rad/s² B. 2 rad/s²
C. 1 rad/s² D. -2 rad/s²

P10. A wheel rotates at frequency 4 Hz. Calculate its angular velocity.

A. 50.27 rad/s B. 12.57 rad/s
C. 25.13 rad/s D. -25.13 rad/s

P11. Define angular displacement.

P12. Define average angular velocity.

P13. Define average angular acceleration.

P14. How are tangential and angular acceleration related for a fixed radius?

P15. Explain why radians should be used in $v = r\omega$.

P16. Convert 135° to radians.

P17. Convert 180° to radians.

P18. A wheel turns through 12 rad in 5 s. Calculate its average angular velocity.

P19. A wheel turns through 16 rad in 6 s. Calculate its average angular velocity.

P20. A point is 0.3 m from a rotating axle with angular velocity 5.0 rad/s. Calculate its tangential speed.

P21. A point is 0.4 m from a rotating axle with angular velocity 5.0 rad/s. Calculate its tangential speed.

P22. A wheel's angular velocity changes from 3 to 21 rad/s in 9 s. Find its average angular acceleration.

P23. A wheel's angular velocity changes from 4 to 28 rad/s in 12 s. Find its average angular acceleration.

P24. A wheel rotates at frequency 6 Hz. Calculate its angular velocity.

P25. A wheel rotates at frequency 8 Hz. Calculate its angular velocity.

P26. Two points on a rigid wheel are at radii r and 2r. Compare their angular velocities and tangential speeds.

P27. A wheel rotates at constant nonzero angular velocity. Is every point's linear velocity constant? Explain.

P28. A disk turns 5 revolutions in 2.0 s at a steady rate. Find angular velocity and the speed of a point 0.10 m from the axis.

P29. Why do different points on Earth travel different distances in a day although they complete the same rotation angle?

P30. Design a simple way to measure the angular velocity of a slow wheel.